



Sericulture

Introduction to Sericulture

- **Sericulture** or **silk production** is the breeding and management of silk worms for the commercial production of silk.
- Rearing of the silkworms on mulberry plants
- Collection of silkworm
- Processing of silkworm cocoons to extract raw silk fibers
- The production of commercial silk.



Silkworms

- The silkworm is the larva or caterpillar of the domesticated silk moth, *Bombyx mori* (Latin: “silkworm of the mulberry tree”).
- It is an economically important insect
- Monophagous
- Selective breeding





Classification

- **Kingdom:** Animalia
- **Phylum:** Arthropoda
- **Class:** Insecta
- **Order:** Lepidoptera
- **Family:** Bombycidae
- **Genus:** Bombyx, Antherea, Phylosamia

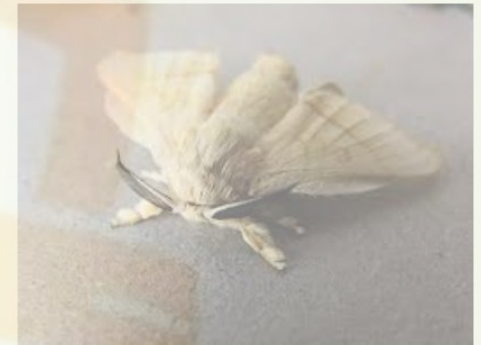
Types of Silkworms:

- There are five major types of silk of commercial importance, obtained from different species of silkworms
- *Bombyx mori meridionalis*, the Mulberry silk worm
- *Anthereapaphia* or *Anthereamylitta*, the Tasar silk worm
- *Antherearoyeli* and *Anthereaperniyi*, the Oak Tasar silk worms
- *Anthereaassama*, the Muga silk worm
- *Phylosamiaricini*, the Eri silkworm

Mulberry:

Bombyx mori meridionalis, the Mulberry silk worm

- Feeds on the leaves
- Commercial silk producer
- Completely domesticated



Anthereapaphia or *Anthereamylitta*, the Tasar silk worm

- Coarse silk
- Less lustrous than mulberry silk
- Reared in nature on the trees



***Antherearoyeli* and *Anthereaperniyi*, the Oak tasar silk worms**

- Introduced from foreign countries
- It feed on natural food plants of oak
- Main producer of Oak Tasar
- Multivoltine



***Anthereaassama*, the Muga silk worm**

- Golden yellow colored silk
- Semi-domesticated
- Multivoltine
- Feed on the aromatic leaves



Phylosamiaricini, the Eri silkworm

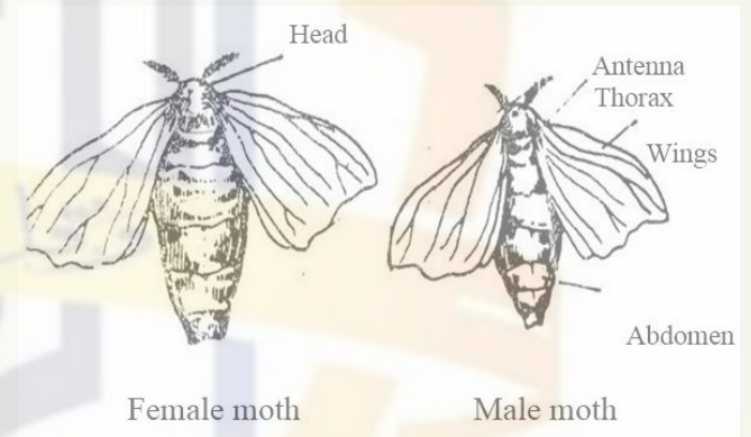
- Feeds on castor
- Multivoltine
- Domesticated



External Features of Mulberry Silkworm

- Size
- Colour
- Body

- Head
- Thorax
- Abdomen
- Compound eyes
- Feathery antennae and the mouth parts.
- Legs and two pairs of wings.
- Scales.



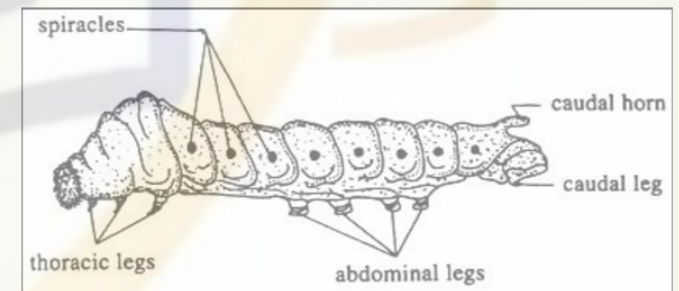
Eggs

- 300 to 400 eggs
- Clusters on the leaves of mulberry tree.
- Gelatinous secretion
- Univoltine
- Multivoltine



Larva

- Mandibulate mouth
- Spinneret
- Thorax
- legs
- Abdomen
- pro-legs or pseudo-legs.
- Dorsal anal horn
- Respiratory spiracles
- voracious eater
- Salivary glands



Pupa

- Feed termination
- Secretion of sticky fluid
- Solid thread/filament formation
- Progressive changes



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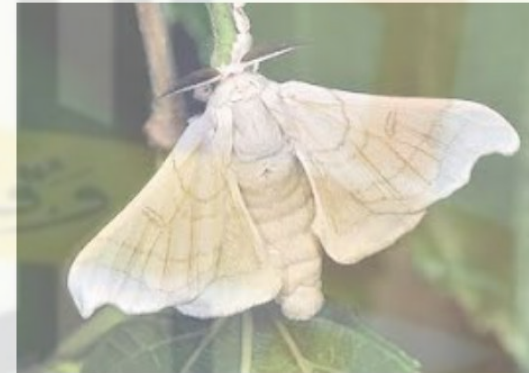
Cocoon

- A single long continuous thread
- Irregular filaments
- Continuous filaments
- Dormant
- Pupa chrysalis
- Metamorphosis
- Regressive Changes



Adult: imago

- Secretion
- Emergence
- Mating
- Egg laying
- Death



History of sericulture

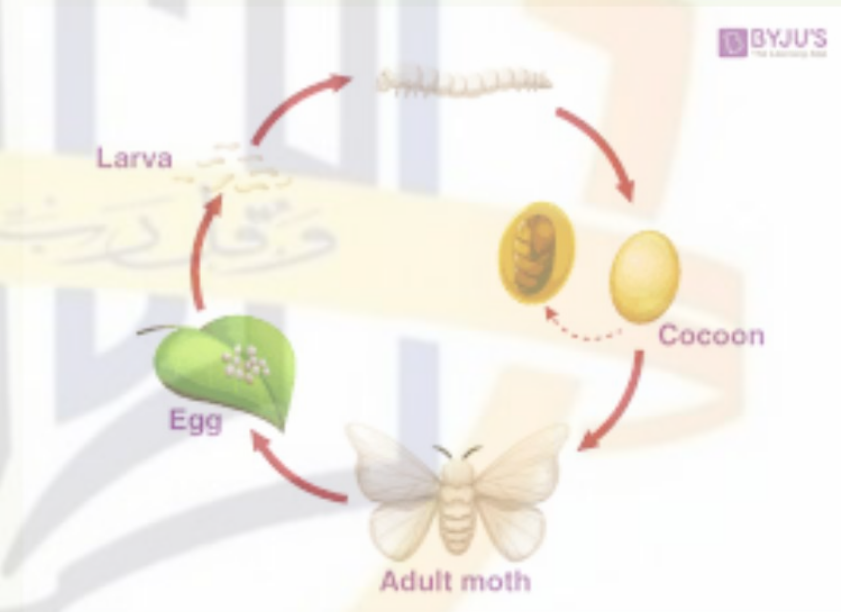
- Silk production from *B. mori* about 2700 BC
- Empress Si-lung-Chi, Start the Silk Industry
- The Goddess of Silk Worm
- Subsequent spread of sericulture
- Silk Route
- Unknown history
- Only living species of family Bombycidae



Life Cycle of Silkworm

These stages are

- Eggs
- Larvae
- Pupae
- Adult (imago)





Cont...

- The whole life cycle completes in five to seven weeks and it depends upon climatic conditions.
- The larvae feed on fresh mulberry leaves.
- The 5th instar larvae feed voraciously
- Spinning of cocoon
- Pupation
- Emergence of Adult.
- The adults don't feed and their only work is to mate.
- They die after few days
- The adults also cannot fly but they have strong smelling power



Importance of Sericulture

- High Employment Potential
- Provides Vibrancy to Rural Economy
- Low Gestation, High Returns
- Women-friendly Occupation
- Ideal Programme for the Weaker Section of the Society
- Eco-Friendly Properties
 - Soil preservation
 - Manures
 - Fuel source
 - Pollution control
 - Integrated farming



Requirements of Sericulture: (Rearing of Silkworm)

- House
- Feeding Trays
- Machan
- Nets
- Spinning Trays
- Chopping knives
- Hygrometer
- Thermometer

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House

- Any building or thatch which is well ventilated may be used for rearing the worms, but mud walled thatched houses are the best as they are cool in summer and warm in winter season
- During summer season water may be sprinkled in thatched to lower the temperature. While during winter season they act like an insulation sheet.



Hygrometer/Thermometer

- For proper growth and development of silkworms the temperature inside the house should be maintained more or less between 70° –75° F with similar percentage of humidity. For this purpose, Thermometer and Hygrometer are used.



Chopping Knives

- The monophagous Mulberry Silkworms prefer to consume Mulberry leaves so, chopping knives are used to cut the leaves to feed the developing larvae.



Feeding Trays

- Freshly hatched worms are kept in flat trays along with small pieces of mulberry leaves. These trays are made up of bamboo mattings with their edges turn up. This type of design may be made up of wood or plastics. The margins should be elevated. On the back of the tray two strong stripes are firmly fastened longitudinally.



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Machan

- Machans are vertical stands made up of bamboos needed to accommodate large number of trays in a limited space. In this way there is no issue of darkness and well ventilated area also available. They are tie across the bars of bamboos in horizontal position. So hence in a limited area a bulk production of silkworm can be done easily.



Nets

Nets are used to prevent Large amount of

- Larval Excreta
- Dirty products
- Remains of leaves



Spinning Trays

- Before cocoon formation, mature worms are transferred to special type of spinning trays known as spinning trays or chandraki.
- Here they spin the cocoon without any disturbance.





Availability of wild flora

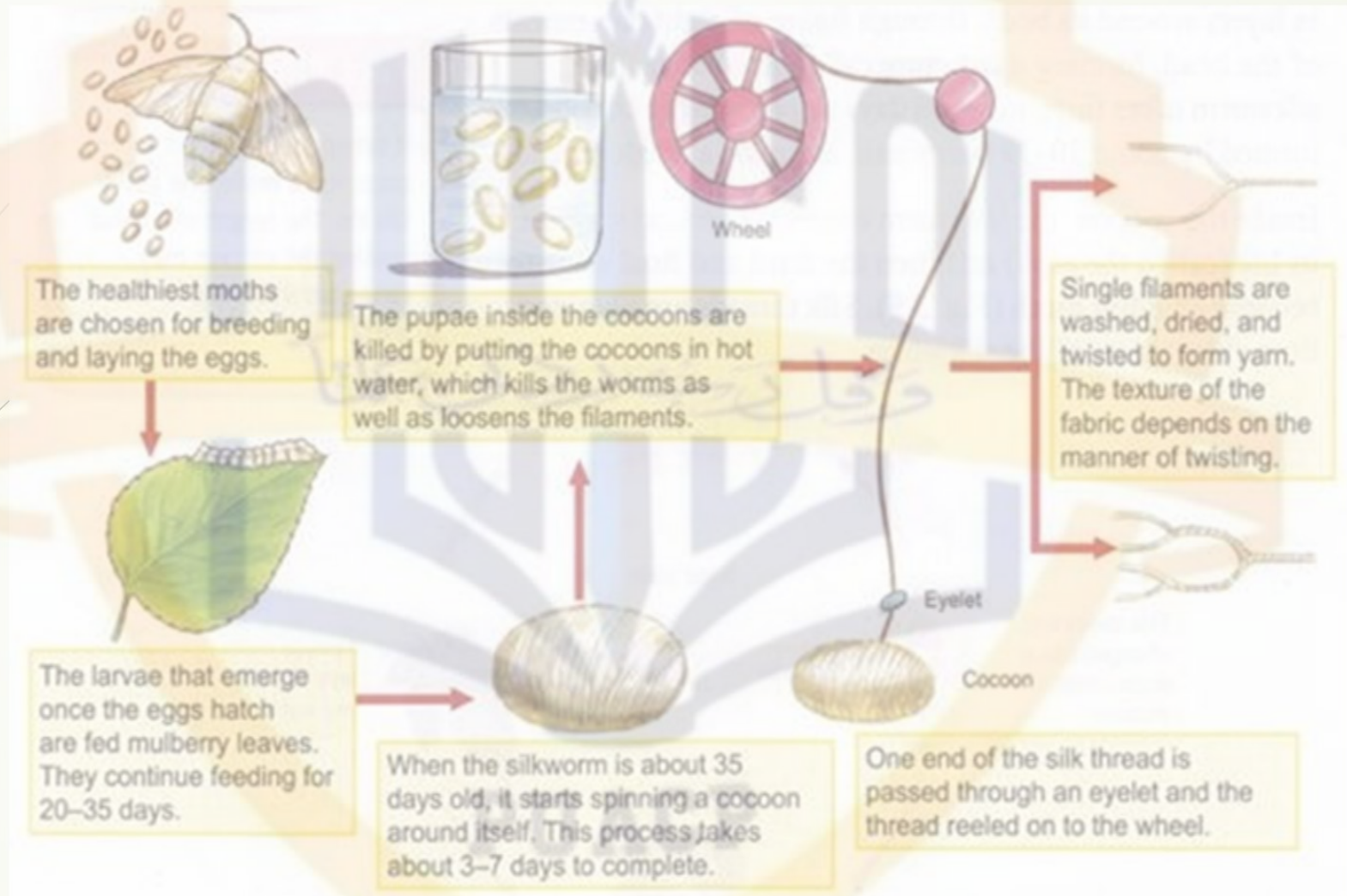
- Plantation of food plants
- Selection of land for cultivation
- Distance between two samplings
- Care of food plants
- Protection and cultivation of food plants:
- Pruning of bushes
- Division of planted plot
- Alternate rearing of silkworms
- Proper care of host plants



Processing of Silk from Cocoon

- Reeling
- Throwing
- Spinning
- Degumming
- Bleaching
- Dyeing
- Printing
- Finishing

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Climatic factors for Sericulture

- Effect of carbon dioxide
- Effect of light
- Effect of temperature

Prospects of Sericulture

The production process of silk fabric contains a long chain of interdependent operations i.e.,

- cultivation of mulberry
- silkworm seed production
- silkworm rearing
- silk production

Problems of Sericulture Industry

- It is hindered by various factors like
- Imports of cheap and alternative textiles from other Asian neighbors
- Use of outdated manufacturing technology
- Primitive and unscientific "reeling/weaving" techniques
- Use of poor quality /diseased seeds
- Low production of bivoltine seeds
- Poor knowledge of farm disease amongst farmers
- Poor supply chain management
- Huge unorganized and decentralized sector
- High production cost



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Recurring droughts and increased import of silk from China accompanied with the following problems like

- Price fluctuation
- Absence of proper market
- Long distance to market
- Lack of transport facilities
- Absence of storage facilities
- Poor information of market trend
- Lack of finance

Diseases of Silkworms

Silkworms suffer from several diseases. Such as

- **Pebrine** caused by a protozoan parasite Nosema.
- The eggs may be discarded or retained according to the presence or absence of parasites.
- **Fletcherie Grasserie jaundice disease**, i.e.,
- losing appetite, showing irregular growths, etc.





Sericulture in Pakistan

- Small industry
- Employment to 25 percent population of rural areas
- Still 15000 families are involved and earning through sericulture
- This can also a source of foreign exchange
- Low investment with high income
- Rearing of Bivoltine species
- Rare Mulberry trees

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